DEBRIEF · SIMULATED CASE

Malignant hyperthermia

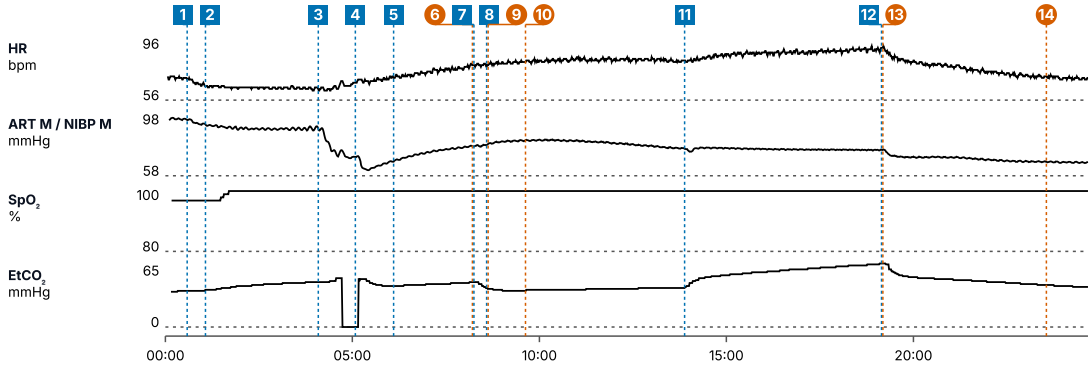
M 35 y 80 kg · 24:40 simulated (2 min 00 s on the clock) · 10/29/2026, 3:26:17 PM · Run by: Live demo, 3 November

Key times

Recognition (instructor's tick)	Start → first team action	Call the MH crisis (stop sevoflurane, charcoal filters...	Give dantrolene 2.5 mg/kg within 300 s of calling the...
2 s 08:12 early signs → 08:14 ticked	**35 s** 00:00 → 00:35 (Midazolam 2 mg and fentanyl 100 µg given, breathing room air)	**2 s** 08:12 → 08:14 · asked ≤ 2 min 00 s	**21 s** 08:14 → 08:35 · asked ≤ 5 min 00 s

The course

■ team action · ● patient event · □ scenario state · numbers match Key moments



Objectives

What the record shows: it reads the run, it does not grade the person

1. ✓ **Observed** Name malignant hyperthermia from the monitor: $EtCO_2$ rising at unchanged ventilation when it should be falling (42 → 45; without MH it settles 36–39) with a rising heart rate (64 → 80, blunted by fentanyl) (instructor-assessed)
   Observed by the instructor at 08:14.
2. ✓ **Observed** Call the MH crisis (stop sevoflurane, charcoal filters, propofol infusion, O_2 100 % at 10 L/min, hyperventilate) within 120 s of the early signs ($EtCO_2$ past 45 mmHg at unchanged ventilation) 2 s of 2 min 00 s
   The state "Trigger off: sevoflurane stopped, charcoal filters, propofol 100 µg/kg/min, O_2 100 % at 10 L/min, ventilation 20 × 600" 2 s after the state "Early signs: $EtCO_2$ rising at unchanged ventilation, heart rate rising" (08:12 → 08:14; within 2 min 00 s asked).
3. ✓ **Observed** Give dantrolene 2.5 mg/kg within 300 s of calling the crisis
   21 s of 5 min 00 s
   Dantrolene 21 s after the state "Trigger off: sevoflurane stopped, charcoal filters, propofol 100 µg/kg/min, O_2 100 % at 10 L/min, ventilation 20 × 600" (08:14 → 08:35; within 5 min 00 s asked).
4. ✓ **Observed** The $EtCO_2$ is climbing back at normal ventilation: the MH is not controlled — second dose of dantrolene 2.5 mg/kg and hyperventilate again (20 × 600, O_2 100 %)
   Dantrolene after the state "Ventilation back to 12 × 500, O_2 100 %" at 19:08
5. – Not assessable Treat the potassium (calcium chloride, insulin with dextrose) within 300 s of K^+ rising above 5.5 mmol/L
   K^+ above 5.5 mmol/L did not happen.
6. ✓ **Observed** Hyperventilation kept until $EtCO_2$ has stayed below 40 mmHg for 2 min, then ventilation normalised (back to 12 × 500)
   The state "Ventilation back to 12 × 500, O_2 100 %" at 13:52, after $EtCO_2$ had stayed below 40 mmHg for 2 min 00 s (08:37 → 10:37).

Key moments and what the monitor showed

values at that second

n	Time	Who	What happened	HR	MAP	SpO_2	$EtCO_2$
1	00:35	Team	Midazolam 2 mg and fentanyl 100 µg given, breathing room air	69	94	96	37
2	01:04	Team	Preoxygenating: 100 % O_2 by face mask	65	91	96	38
3	04:05	Team	Induced: propofol 2 mg/kg, succinylcholine 1.5 mg/kg (masseter rigidity); airway not yet secured	63	89	99	46
	Alarms since the last row: RR low, RR low						
4	05:04	Team	Intubated: ventilator 12 × 500, PEEP 5, FiO_2 0.5; sevoflurane 2 % at 2 L/min	67	70	99	0
	Alarms since the last row: RR low, $EtCO_2$ low, Apnoea (no CO_2 breaths)						
5	06:06	Team	Surgery under way: sevoflurane 2 %, rocuronium	70	67	99	42
	Alarms since the last row: RR low, awRR low, NIBP S low						
6	08:12	Patient	Early signs: $EtCO_2$ rising at unchanged ventilation, heart rate rising	78	77	99	46
7	08:14	Team	Trigger off: sevoflurane stopped, charcoal filters, propofol 100 µg/kg/min, O_2 100 % at 10 L/min, ventilation 20 × 600	78	77	99	46
8	08:35	Team	Dantrolene 2.5 mg/kg given	77	78	99	40
9	08:37	Patient	MH treatment under way	78	78	99	39
10	09:37	Patient	Recovering: $EtCO_2$ below 45	80	80	99	37
11	13:52	Team	Ventilation back to 12 × 500, O_2 100 %	81	76	99	40
12	19:08	Team	Dantrolene 2.5 mg/kg again; hyperventilation again: 20 × 600, O_2 100 %	89	75	99	65
	Alarms since the last row: $EtCO_2$ high						
13	19:11	Patient	MH treatment under way	89	75	99	65
14	23:33	Patient	Recovering: $EtCO_2$ below 45	71	67	99	43
	Alarms since the last row: NIBP S low						

Measured facts

Lowest SpO_2	96 % at 00:10
Lowest ART M / NIBP M	62 mmHg at 05:25
Highest $EtCO_2$	65 mmHg at 19:06
Highest T1 / T2	36.8 °C at 00:01
Apnoea alarm to ventilated breaths	18 s (04:47 → 05:04)
Labs (the model): last (lowest–highest)	K^+ 5.0 mmol/L (4.2–5.0) · Lactate 3.6 mmol/L (1.0–3.6) · pH 7.33 (7.18–7.41) · Glucose 6.6 mmol/L (5.6–6.6) · Tcore 36.4 °C (36.4–36.8)

Talking points

Reactions
How did that feel? What stands out for you?

Description
In a sentence or two: what was the problem, and when did you know?

Analysis
Take one key moment: what did you see, what did you think, what did you do?

Summary
One thing to keep doing, one thing to do differently next time.

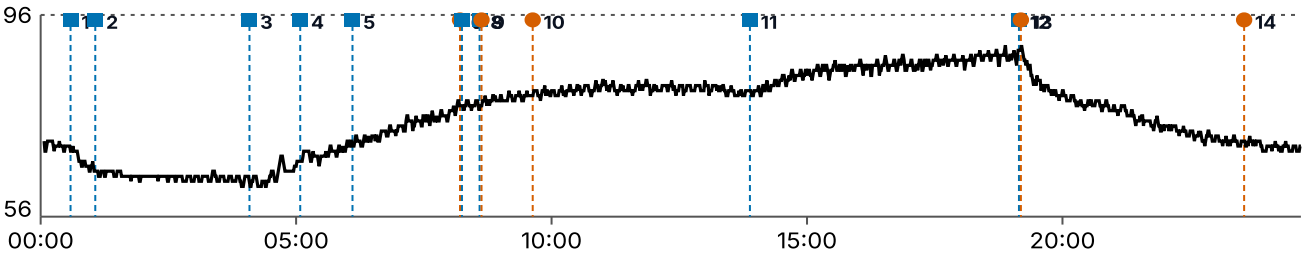
Notes

Kit round A4 rehearsal of the live-demo path.

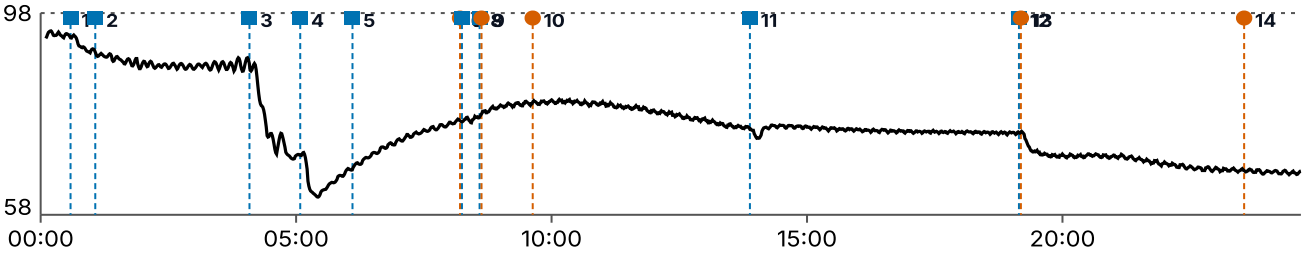
Trends

■ team action · ● patient event · | scenario state

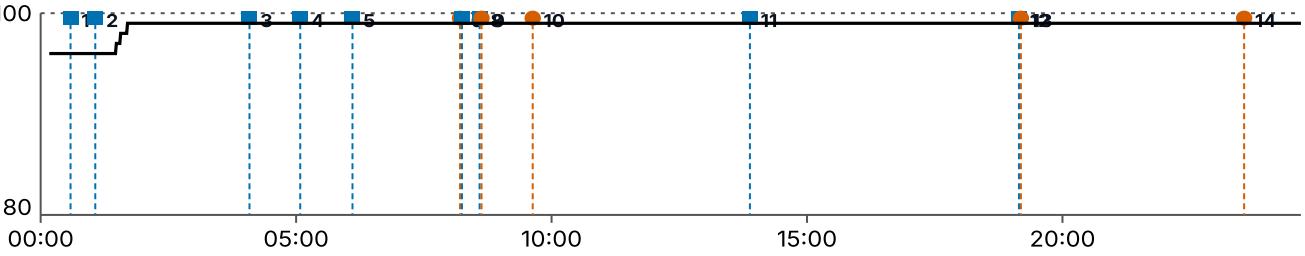
HR bpm



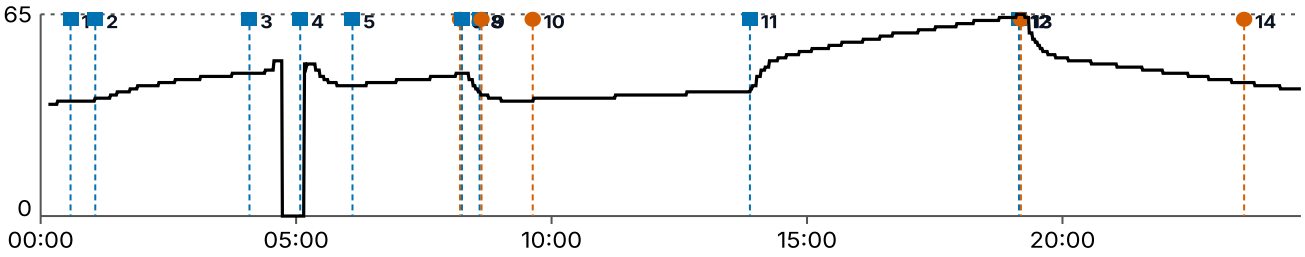
ART M / NIBP M mmHg



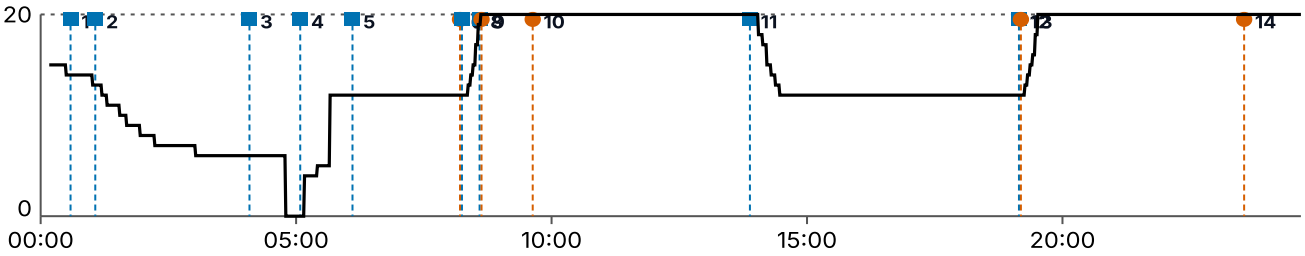
SpO_2 %



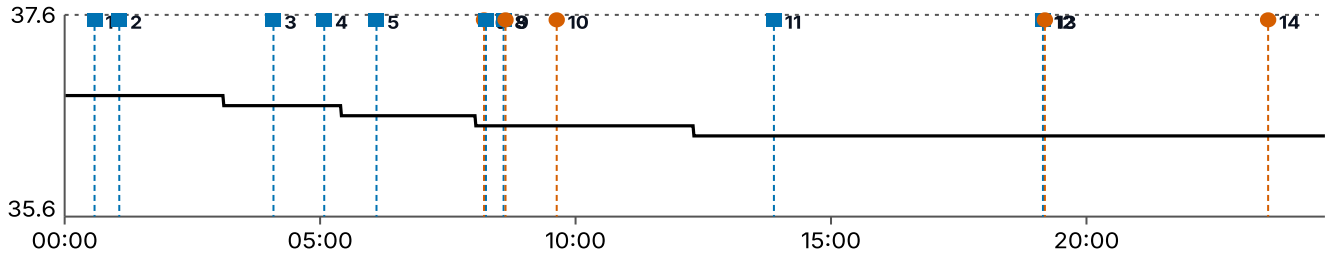
$EtCO_2$ mmHg



awRR / RR /min



T1 / T2 °C



Drugs and fluids

Time	Given
00:35	Midazolam 2 mg IV
00:35	Fentanyl 100 µg IV
04:05	Propofol 2 mg/kg IV
04:05	Succinylcholine 1.5 mg/kg IV
06:06	Rocuronium 0.6 mg/kg IV
08:14	Propofol 100 µg/kg/min infusion started
08:35	Dantrolene 2.5 mg/kg IV
19:08	Dantrolene 2.5 mg/kg IV

- Midazolam: 2 mg in 1 bolus
- Fentanyl: 100 µg in 1 bolus
- Propofol: 160 mg in 1 bolus
- Succinylcholine: 120 mg in 1 bolus
- Rocuronium: 48 mg in 1 bolus
- Dantrolene: 400 mg in 2 boluses

Timeline

Time	Kind	Event
00:00	Scenario	Scenario loaded: Malignant hyperthermia
00:00	Scenario	Scenario state: Awake on the table, monitoring on, breathing room air
00:01	Action	Speed ×8
00:04	Bookmark	Bookmark: Bookmark 1 at 00:04
00:35	Scenario	Scenario state: Midazolam 2 mg and fentanyl 100 µg given, breathing room air
00:35	Scenario	Midazolam 2 mg IV
00:35	Scenario	Fentanyl 100 µg IV
01:04	Scenario	Scenario state: Preoxygenating: 100 % O_2 by face mask
01:04	Scenario	Preoxygenation, FiO_2 100 %
03:17	Alarm	RR low
04:04	Alarm	RR low
04:05	Scenario	Scenario state: Induced: propofol 2 mg/kg, succinylcholine 1.5 mg/kg (masseter rigidity); airway not yet secured
04:05	Scenario	Propofol 2 mg/kg IV

Time	Kind	Event
04:05	Scenario	Succinylcholine 1.5 mg/kg IV
04:05	Scenario	Malignant hyperthermia: severe
04:34	Alarm	RR low
04:45	Alarm	$EtCO_2$ low
04:47	Alarm	Apnoea (no CO_2 breaths)
05:04	Scenario	Scenario state: Intubated: ventilator 12 × 500, PEEP 5, FiO_2 0.5; sevoflurane 2 % at 2 L/min
05:04	Scenario	Tracheal tube in place
05:04	Scenario	Preoxygenation, FiO_2 21 %
05:04	Scenario	Ventilator: RR 12 /min, VT 500 mL, PEEP 5 cmH_2O, FiO_2 50 %
05:04	Scenario	Sevoflurane 2 % at 2 L/min fresh gas
05:04	Patient	Ventilated breaths began
05:10	Alarm	RR low
05:10	Alarm	awRR low
05:35	Alarm	NIBP S low
06:06	Scenario	Scenario state: Surgery under way: sevoflurane 2 %, rocuronium
06:06	Scenario	Rocuronium 0.6 mg/kg IV
07:34	Action	Speed ×4
08:12	Scenario	Scenario state: Early signs: $EtCO_2$ rising at unchanged ventilation, heart rate rising
08:14	Note	Objective met: Name malignant hyperthermia from the monitor: $EtCO_2$ rising at unchanged ventilation when it should be falling (42 → 45; without MH it settles 36–39) with a rising heart rate (64 → 80, blunted by fentanyl) (instructor-assessed)
08:14	Scenario	Scenario state: Trigger off: sevoflurane stopped, charcoal filters, propofol 100 µg/kg/min, O_2 100 % at 10 L/min, ventilation 20 × 600
08:14	Scenario	Sevoflurane vaporiser off
08:14	Scenario	Propofol 100 µg/kg/min infusion started
08:14	Scenario	Ventilator: RR 20 /min, VT 600 mL, PEEP 5 cmH_2O, FiO_2 100 %
08:35	Scenario	Scenario state: Dantrolene 2.5 mg/kg given
08:35	Scenario	Dantrolene 2.5 mg/kg IV
08:37	Scenario	Scenario state: MH treatment under way
08:38	Action	Fast-forward 5 min
09:37	Scenario	Scenario state: Recovering: $EtCO_2$ below 45
13:52	Scenario	Scenario state: Ventilation back to 12 × 500, O_2 100 %
13:52	Scenario	Ventilator: RR 12 /min, VT 500 mL, PEEP 5 cmH_2O, FiO_2 100 %
13:56	Action	Fast-forward 5 min
14:27	Alarm	$EtCO_2$ high
19:08	Scenario	Scenario state: Dantrolene 2.5 mg/kg again; hyperventilation again: 20 × 600, O_2 100 %
19:08	Scenario	Dantrolene 2.5 mg/kg IV
19:09	Scenario	Ventilator: RR 20 /min, VT 600 mL, PEEP 5 cmH_2O, FiO_2 100 %
19:11	Scenario	Scenario state: MH treatment under way
19:12	Action	Fast-forward 5 min

Time	Kind	Event
20:28	Alarm	NIBP S low
23:33	Scenario	Scenario state: Recovering: $EtCO_2$ below 45
24:26	Action	Simulation paused
24:26	Action	Simulation resumed

Bookmarks

- 00:04 Bookmark: Bookmark 1 at 00:04

Appendix: one row per minute

Time	HR bpm	ART M / NIBP M mmHg	SpO_2 %	$EtCO_2$ mmHg	awRR / RR /min	T1 / T2 °C	K^+ mmol/L	Lactate mmol/L	pH	Glucose mmol/L	Tcore °C
00:00	71	93	96	36	15	36.8	4.2	1.0	7.41	5.6	36.8
01:00	66	91	96	37	14	36.8	4.2	1.0	7.39	5.6	36.8
02:00	64	89	99	42	8	36.8	4.2	1.0	7.36	5.6	36.8
03:00	63	87	99	44	7	36.8	4.2	1.0	7.33	5.6	36.7
04:00	64	88	99	46	6	36.7	4.2	1.0	7.32	5.6	36.7
05:00	66	70	99	0	0	36.7	4.3	1.0	7.27	5.6	36.6
06:00	71	67	99	42	12	36.6	4.5	1.0	7.34	5.6	36.6
07:00	74	73	99	44	12	36.6	4.6	1.0	7.34	5.6	36.6
08:00	77	77	99	45	12	36.6	4.8	1.0	7.32	5.6	36.5
09:00	80	80	99	38	20	36.5	4.7	1.1	7.39	5.7	36.5
10:00	81	81	99	38	20	36.5	4.7	1.2	7.39	5.8	36.5
11:00	83	81	99	38	20	36.5	4.6	1.3	7.38	5.8	36.5
12:00	82	79	99	39	20	36.5	4.6	1.5	7.37	5.9	36.4
13:00	83	78	99	40	20	36.4	4.6	1.6	7.36	6.0	36.4
14:00	81	74	99	42	20	36.4	4.6	1.8	7.32	6.0	36.4
15:00	84	76	99	53	12	36.4	4.7	2.0	7.26	6.1	36.4
16:00	86	75	99	56	12	36.4	4.7	2.2	7.24	6.2	36.4
17:00	86	75	99	59	12	36.4	4.7	2.4	7.22	6.2	36.4
18:00	87	75	99	62	12	36.4	4.8	2.6	7.20	6.3	36.4
19:00	87	75	99	64	12	36.4	4.9	2.8	7.18	6.4	36.4
20:00	81	71	99	51	20	36.4	4.9	3.0	7.26	6.4	36.4
21:00	78	70	99	49	20	36.4	4.9	3.2	7.27	6.5	36.4
22:00	74	69	99	46	20	36.4	4.9	3.3	7.29	6.5	36.4
23:00	72	68	99	44	20	36.4	5.0	3.5	7.30	6.5	36.4
24:00	70	67	99	42	20	36.4	5.0	3.6	7.32	6.6	36.4